NanoGT Match Report: GATA2 deficiency

Disease: GATA2 deficiency (ORPHA:247770)
Primary gene: GATA2
Gene CDS: 1443 bp
Inheritance: Autosomal dominant
Target tissues scored: hematopoietic

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility; Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology; Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable.

Disease Mechanism Evidence

Molecular mechanism: haploinsufficiency
Mechanistic detail: Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema
Gene-addition compatibility: conditional
Preferred modality class: ex vivo hsc gata2 gene addition
Evidence level/status: direct / source_checked
Evidence summary: Spinner MA et al. (2014 Blood PMID 24227816) characterised the full clinical spectrum of GATA2 haploinsufficiency; gene addition is conditional via ex vivo HSC transduction because systemic in vivo delivery risks ectopic GATA2 expression in non-haematopoietic tissues where dosage must be precisely regulated
Evidence source: Spinner MA et al. 2014 Blood PMID 24227816

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Few catalog vectors cover the annotated disease tissue(s): LV.
- Endpoint readiness unclear from available annotations; identify biomarkers, natural-history measures, and patient-relevant outcomes before trial prioritisation.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Strimvelis	LV	7.6/10	High	approved

Rank	Program	Vector	Score	Confidence	Approval
2	Libmeldy	LV	6.4/10	Medium	approved
3	Skysona	LV	6.4/10	Medium	approved
4	AVR-RD-01	LV	6.1/10	Medium	phase1/2
5	BMN 307	AAV5	5.6/10	Medium	phase2

Match #1: Strimvelis

Precedent disease: ADA-SCID
Vector: LV
Tissue target: hematopoietic
Composite score: 7.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.57**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1443bp / cargo 8000bp (18% utilized)

- Vector tropism plus precedent target match: hematopoietic
- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Pathway match: immune_hematopoietic
- Disease mechanism: haploinsufficiency — Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Spinner MA et al. (2014 Blood PMID 24227816) characterised the full clinical spectrum of GATA2 haploinsufficiency; gene addition is conditional via ex vivo HSC transduction because systemic in vivo delivery risks ectopic GATA2 expression in non-haematopoietic tissues where dosage must be precisely regulated
- Mechanism source: Spinner MA et al. 2014 Blood PMID 24227816 (https://pubmed.ncbi.nlm.nih.gov/24227816/
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — GATA2 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #2: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 6.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.38**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1443bp / cargo 8000bp (18% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (immune_hematopoietic vs leukodystrophy)
- Disease mechanism: haploinsufficiency — Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Spinner MA et al. (2014 Blood PMID 24227816) characterised the full clinical spectrum of GATA2 haploinsufficiency; gene addition is conditional via ex vivo HSC transduction because systemic in vivo delivery risks ectopic GATA2 expression in non-haematopoietic tissues where dosage must be precisely regulated
- Mechanism source: Spinner MA et al. 2014 Blood PMID 24227816 (https://pubmed.ncbi.nlm.nih.gov/24227816/
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate

neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — GATA2 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #3: Skysona

Precedent disease: Cerebral adrenoleukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 6.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.38**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1443bp / cargo 8000bp (18% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (immune_hematopoietic vs peroxisomal)
- Disease mechanism: haploinsufficiency — Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Spinner MA et al. (2014 Blood PMID 24227816) characterised the full clinical spectrum of GATA2 haploinsufficiency; gene addition is conditional via ex vivo HSC transduction because systemic in vivo delivery risks ectopic GATA2 expression in non-haematopoietic tissues where dosage must be precisely regulated
- Mechanism source: Spinner MA et al. 2014 Blood PMID 24227816 (https://pubmed.ncbi.nlm.nih.gov/24227816/
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — GATA2 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #4: AVR-RD-01

Precedent disease: Fabry disease
Vector: LV
Tissue target: hematopoietic
Composite score: 6.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**6.14**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1443bp / cargo 8000bp (18% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (immune_hematopoietic vs lysosomal_storage)
- Disease mechanism: haploinsufficiency — Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Spinner MA et al. (2014 Blood PMID 24227816) characterised the full clinical spectrum of GATA2 haploinsufficiency; gene addition is conditional via ex vivo HSC transduction because systemic in vivo delivery risks ectopic GATA2 expression in non-haematopoietic tissues where dosage must be precisely regulated
- Mechanism source: Spinner MA et al. 2014 Blood PMID 24227816 (https://pubmed.ncbi.nlm.nih.gov/24227816/
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — GATA2 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #5: BMN 307

Precedent disease: Phenylketonuria
Vector: AAV5
Tissue target: liver
Composite score: 5.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**5.62**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1443bp / cargo 4700bp (31% utilized)
- No tissue overlap (disease: [‘hematopoietic’], vector: [‘liver’, ‘lung’, ‘CNS’])
- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Different pathway (immune_hematopoietic vs amino_acid_metabolism)
- Disease mechanism: haploinsufficiency — Autosomal dominant haploinsufficiency of GATA2 transcription factor depletes haematopoietic stem cells and lymphatic endothelial progenitors causing combined immunodeficiency bone marrow failure and lymphedema
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Spinner MA et al. (2014 Blood PMID 24227816) characterised the full clinical spectrum of GATA2 haploinsufficiency; gene addition is conditional via ex vivo HSC transduction because systemic in vivo delivery risks ectopic GATA2 expression in non-haematopoietic tissues where dosage must be precisely regulated
- Mechanism source: Spinner MA et al. 2014 Blood PMID 24227816 (https://pubmed.ncbi.nlm.nih.gov/24227816/
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden

- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders
- ORGANELLE TARGETING: COMPATIBLE — GATA2 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: hematopoietic
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone